Software Requirements Specification (SRS)

Project Name: Gradify

1. Introduction

Purpose:

Gradify automates evaluation of student answer sheets using OCR, answer matching, and CSV-based scoring export.

Scope:

- Upload answer key

- Upload student answer sheets (webcam or file)

- OCR extraction

- Automated scoring

- CSV generation

Tech Stack: Next.js, Node.js, Express, MongoDB, Tesseract.js OCR

2. Overall Description

Stakeholders: Faculty/Admin users

System Overview: OCR pipeline, evaluation engine, data storage, reporting

3. Functional Requirements

FR1: Answer key upload

FR2: Student answer sheet upload

FR3: OCR extraction of text + student info

FR4: Mapping extracted answers to key

FR5: Automated scoring (question-wise + total)

FR6: CSV report generation

FR7: Results dashboard

4. Non■Functional Requirements

Performance: OCR per sheet < 10 seconds

Usability: Simple UI with guided steps

Security: Temporary data storage, auto-delete

Reliability: 50+ sheets batch processing

Maintainability: Modular backend + reusable frontend components

5. Cost Breakdown

Frontend (Next.js UI) – ■2,500

Backend (Node.js + Express APIs) – ■4,700

Database Layer (MongoDB Atlas) – ■1,700

OCR Processing (Tesseract.js Integration) – ■2,300

Evaluation Engine – ■1,600

CSV Export Module – ■1,000

Deployment & Hosting (Optional domain) – ■700

Documentation + Testing – ■1,500

Total Estimated Cost: ■15,000

6. Payment Terms

25% advance before starting the project.

75% payable after successful completion & delivery of all modules.

7. Conclusion

Gradify provides a fast, automated, cost-effective exam evaluation solution using a modern and budget-friendly tech stack. The system is scalable, secure, and optimized for student and institutional workflows.